

Insights from the Stack Overflow Developer Survey 2024: Technology Trends, Skill Demand, and Salary Patterns

Survey of 65,000+ professional developers across 314 technologies. Salary data normalised to USD per year.

1. Most-Used Technologies

Among professional developers, JavaScript remains the most used language (63.6%), followed by HTML/CSS (52.8%), Python (51.0%), SQL (49.3%), and TypeScript (38.5%). Java is reported by 30.3% of respondents, Bash/Shell by 27.2%, and C# by 27.1%. Go, Rust, Kotlin, and Scala remain niche but growing — under 12% adoption each, but consistently linked to higher compensation.

On the cloud side, AWS leads the market at 48% adoption, followed by Azure (28%) and Google Cloud Platform (GCP) at 25%. Kubernetes and Docker dominate container orchestration — 22% and 53% adoption respectively. Frameworks like Node.js (40.8%), React (39.5%), Spring (15.0%), Django (13.4%), and FastAPI (9.9%) round out the backend / frontend stack.

2. Highest Paying Programming Languages

Language	Median Salary (USD/yr)	Top Region
Erlang	\$108,418	United States
Elixir	\$96,387	United States
Clojure	\$95,701	United States
Lisp	\$95,000	United States
Ruby	\$92,500	United States
Rust	\$89,847	United States
F#	\$86,503	United Kingdom
Scala	\$85,000	Germany
Go	\$83,750	United States
Kotlin	\$80,000	Germany

Table 1. Top 10 highest paying programming languages from the 2024 survey.

3. Role-Based Compensation

Senior executives earn the most on average at approximately \$225,000 USD per year, followed by engineering managers (\$150,000), developer advocates (\$120,000), and developer experience professionals (\$118,500). The middle of the distribution sits around full-stack and back-end developers — \$100,000 to \$115,000. Mobile developers, embedded developers, and academic researchers trail at

\$70,000 to \$85,000.

4. AI Tooling Adoption

Most developers (76%) use AI tools as part of their workflow. ChatGPT is the most popular at 74% adoption, followed by GitHub Copilot. Among developers currently using AI tools, 41% report wanting to use GitHub Copilot in the next year. Other tools like Tabnine, Cursor, and Codeium have smaller but growing user bases. Misinformation in AI results is the top ethical concern for 79% of developers, followed by source attribution at 65%.

5. Work Mode and Compensation

42% of developers report working hybrid, while 38% are fully remote and 20% in-person. The in-person share has increased for the third year in a row (2022: 15%, 2023: 16%, 2024: 20%). Remote developers earn on average 8% more than office-based peers in the same role band, though the geography premium has shrunk by two-thirds since 2019.

6. Case Study: Ruby Developer Challenges

An analysis of 498,719 Ruby-related questions from Stack Overflow (2008-2024) combined with a survey of 154 Ruby developers identified 35 distinct topic areas grouped into six major categories. Web Application Development dominates at 27.55% of all questions, followed by Application Quality and Security at 23.28%, Data Management and Processing at 19.71%, Development Environment and Infrastructure at 15.62%, Core Ruby Concepts at 11.22%, and Software Architecture and Performance at 2.59%.

Despite Ruby's decline in market share — from 6% peak in 2012 to 2% in 2020 — it remains in active use across major platforms. Airbnb, GitHub, Shopify and over 400,000 production websites still run on Ruby on Rails. The most challenging area for Ruby developers remains Core Ruby Concepts: 31.6% of surveyed developers rated metaprogramming and advanced method handling as difficult or very difficult.

7. Top Skills by Demand and Pay

Skill	Demand Rank	Median Salary (USD/yr)
Python	1	\$95,000
JavaScript	2	\$85,000
SQL	3	\$92,000
TypeScript	4	\$92,500
Java	5	\$84,000
AWS	6	\$110,000
Docker	7	\$98,000
Kubernetes	8	\$112,000
Kafka	9	\$128,000
Rust	10	\$89,847
Machine Learning	11	\$120,000
Microservices	12	\$108,000

Table 2. Top demanded skills and their median compensation, 2024.

8. Methodology

The 2024 Stack Overflow Developer Survey collected 65,437 responses from developers in 185 countries between May and June 2024. Compensation responses were normalised to USD per year using mid-year exchange rates. Outliers above \$1,000,000 were trimmed before computing medians. Skill frequencies were derived from the optional 'technologies used' multi-select question. Geographical filtering used country of residence rather than country of employment.